KAMALESH JAYAPANDIARAJ ARUMUGAM

OSCP | Application & Cloud Security Engineer | AppSec, API Security, AWS, IaC, DevSecOps
California, United States | karumuga@usc.edu | +1 (213)-691-4175 | LinkedIn | GitHub | Portfolio

Application and cloud security engineer focused on securing cloud-native applications, REST APIs, and authentication and authorization flows. Hands-on experience building and hardening Go and Python services, embedding security into CI/CD pipelines, and enforcing least-privilege access across AWS, Azure, and multi-cloud environments with Terraform and policy-as-code.

EDUCATION

University of Southern California (USC) — Los Angeles, CA, USA
Master of Science – Cybersecurity, GPA: 3.9/4 — August 2024 – May 2026
Relevant Coursework: INF 519 Foundations & Policy for Information Security, CSCI 530 Security Systems, INF 523 Computer Systems Assurance, ITP 425 Web Application Security, INF 529 Privacy in Informatics, INF 528 Computer and Digital Forensics, INF 525 Trusted System Design

PROFESSIONAL EXPERIENCE

Keck Medicine of USC — Los Angeles, CA, USA
Cloud Security Engineer Intern — October 2024 – May 2026

- Built and secured a **Go REST API (~20 endpoints; Azure AD OAuth2, JWT, RBAC)** powering the Keck Cloud Portal for 40+ cloud, infrastructure, and security engineers, enforcing object-level least-privilege access over ~1,000 S3-hosted documents.
- Discovered and remediated an **IDOR / broken object-level authorization flaw** during internal AppSec testing that could have exposed ~1,000 cross-user onboarding requests, approvals, and attached documents; **added server-side ownership checks and RBAC validation on every object-access endpoint,** closing the gap before production rollout with no known exploitation.
- Co-developed the Keck Cloud Portal, **a security-by-default provisioning platform** (Terraform, AWS Lambda, policy-as-code) adopted by 3-5 teams to onboard ~40 cloud workloads, cutting secure onboarding time from 2-3 days to under 1 day.
- **Automated continuous compliance evaluation** of 100+ AWS resources against HIPAA, HITRUST, CIS, NIST, and CMMC baselines using **Terraform (Infrastructure as Code, IaC) and AWS Config Rules**, reducing manual compliance effort by ~25%.
- Built **Python Lambda automation for Security Hub** finding enrichment, alert routing, and remediation playbooks, handling ~65% of routine cloud security findings and removing ~10 hours/week of manual triage.
- Built a **GPT-powered AI triage assistant** (OpenAI API on AWS) that summarized and prioritized ~75 Security Hub, GuardDuty, and Config findings/day into natural-language remediation guidance, cutting analyst triage time from ~12 to ~4 minutes per finding (~5 hours/week saved).

Oracle Corporation — Bangalore, KA, India
Associate Cloud Engineer – JAPAC — September 2022 – July 2024

- Audited **IAM** across **10+ OCI tenancies** via Cloud Guard, replacing standing admin with **just-in-time privileged access** and remediating over-privileged roles, wildcard policy statements, and weak federation trust configs, **cutting standing privileged access by 90%.**
- Embedded **SAST, IaC validation, and policy-as-code gates** into **12 CI/CD pipelines** across 4 engineering teams, blocking misconfigurations and policy violations pre-merge and **reducing security issues reaching pre-production by 25%.**
- Stood up Cloud Guard detector and responder recipes mapped to the **CIS OCI Benchmark**, driving misconfiguration remediation across 15 compartments and raising conformance from 58% to 96%.
- Hardened **20+ migrated OCI workloads** (Kubernetes, Docker) with NSG least-privilege rules, network segmentation, and centralized secrets/key management **via OCI Vault and KMS**, eliminating hardcoded credentials and enforcing encryption-at-rest across multi-tenant environments.

AWARDS & LEADERSHIP

25th in NAMER - Amazon AppSec CTF: Placed 25/143 in a 48-hr invitational AppSec competition (Secure coding, Web, Cloud, AI) **(Writeup)**
Secured 2nd place in the Tenable × CRACCON 2025 Cloud & IAM CTF – Cloud exposure analysis, CNAPP security, IAM misconfigurations.
Top 10, WRCCDC 2026 - Defended enterprise systems in a live attack scenario; **SIEM, firewall hardening** and performed **Incident Response.**
Offensive Security Team Member, CPTC 2025 - Represented USC in a real-world enterprise penetration testing competition. **(Red-Teaming)**
2nd Runner-Up: USC CTF Competition (Fall 2024) – Solved web security, forensics, reverse engineering & cryptography challenges. **(Writeup)**

PROJECTS

LLM & Agentic Security Red-Team Harness (GitHub)

- Built a **red-team harness** that tests the agentic attack surface of LLM tool-using and multi-agent systems, covering **tool-call injection, indirect/RAG injection, excessive agency, cross-agent trust, and memory poisoning**, with probes mapped to the **OWASP Top 10 for LLM.**
- Demonstrated **capability-scoped tool authorization** enforcing least-privilege at the tool-execution boundary to cap blast radius of any injection.

Authorization Testing Framework for REST APIs (GitHub)

- Built a Go-based test harness that **automates broken-object-level (BOLA/IDOR)** and **broken-function-level authorization** testing against REST APIs, walking object IDs across user contexts to surface horizontal and vertical privilege-escalation gaps, with reusable test cases mapped to OWASP API Security Top 10.

Secure SDLC & DevSecOps Pipeline (GitHub)

- Engineered a **CI/CD-integrated Secure SDLC** using GitHub Actions, Semgrep, SonarQube, ZAP, Burp Suite, and Trivy, automating SAST/DAST/container scanning, SBOM generation, and **policy-as-code** gates for continuous code integrity and software-supply-chain security.

IAM Privilege-Escalation Path Analyzer (GitHub)

- Built a **policy-analysis tool that ingests AWS IAM policies** and flags wildcard permissions, over-permissive trust relationships, and known privilege-escalation chains, codifying the misconfiguration patterns behind real-world cloud IAM compromises.

TECHNICAL SKILLS

Certifications: OSCP | Oracle Cloud Infrastructure Architect Professional | Azure SC – 900 | eJPT | CompTIA Security +
Application Security: API Security, OWASP API Security Top 10, AuthN/AuthZ, BOLA/IDOR, OWASP Top 10, OWASP ASVS, Secure SDLC, SAST/DAST, Container Security, LLM/AI Application Security, Threat Modeling (STRIDE), CWE Top 25, OWASP Top 10 for LLM Applications
Cloud & IaC: AWS (EC2, Lambda, S3, IAM, GuardDuty, Security Hub, KMS, Config), Azure AD, GCP, Terraform, Ansible, Kubernetes, Docker, Zero Trust, CSPM, CNAPP/CWPP
Languages: Python, Go, Bash, PowerShell, SQL, Java, C/C++, JavaScript, YAML, JSON
Tools & Frameworks: Burp Suite, OWASP ZAP, Postman, Semgrep, SonarQube, Trivy, Nmap, GitHub Actions, CIS Benchmarks, CMMC, HIPAA